

MAAHIR RAJESH GARG

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Education

National University of Singapore

Aug 2023 – May 2027 (Expected)

B.Comp. (Honours), Computer Science; Second Major in Quantitative Finance | GPA 4.71 / 5.00

Singapore

- Minor in Data Engineering; Specialisation in Artificial Intelligence (Distinction).
- **A+**: Multivariable Calculus, Quantitative Finance, Discrete Structures, AI & Machine Learning, Foundations of AI; **A**: Algorithms, Data Structures, Regression Analysis (**A–** Probability & Statistics).
- Coursework: Probability & Statistics, Linear Algebra, Stochastic Processes, Regression, Optimisation, Algorithms, Quantitative Finance.

Technical Skills

Quantitative: probability, statistics, regression, linear algebra, multivariable calculus, stochastic processes, optimisation, expected-value reasoning, hypothesis testing

Programming: Python, C/C++, Java, SQL, R, JavaScript/TypeScript, Bash, LaTeX

Data & ML: NumPy, Pandas, scikit-learn, PyTorch, Spark, Kafka, Airflow, PostgreSQL, Snowflake, Docker, AWS, Git

Experience

GIC

Jan 2026 – Present

AI Engineer

Singapore

- Built an internal **agentic CLI and Chainlit assistant** over proprietary LLMs in a sandboxed Docker/AWS environment, with **8 custom tools**, multi-agent orchestration, skill-routing, and step-level guardrails via custom hooks, equipping an **11-person team** where classified-data constraints rule out tools like Copilot.
- Designed an **agentic audit pipeline** that calls Snowflake and SharePoint as tools, runs ETL, and generates audit-ready reports from plain-English prompts; **automated 2 previously manual data sources**, cutting evidence retrieval from **hours to near-instant**.

Interactive 3D Lab

May 2025 – Dec 2025

Swift Developer, Apple Vision Pro

Singapore

- Co-invented a **multimodal tracking framework** (patent application in progress) unifying Apple Vision Pro and iPhone into one tracking surface at **89% cross-device accuracy** (Swift, RealityKit) for hand-eye coordination assessment in stroke rehabilitation; **co-author** on a paper in preparation.
- Shipped the visionOS interaction layer end-to-end: spatial UI and real-time hand/gaze tracking, with an iPhone companion app on a low-latency local channel at **30 fps and sub-millisecond latency**.

NUS, School of Computing

Aug 2024 – Dec 2025

Undergraduate Teaching Assistant (Algorithms, Data Structures, Discrete Math)

Singapore

- TA for CS3230 (Design & Analysis of Algorithms), CS2040S (Data Structures & Algorithms), and CS1231S (Discrete Structures) across three semesters; coached ~20 students through dynamic programming, graph algorithms, and complexity analysis.
- Earned **4.8 / 5.0** teaching effectiveness against a 4.2 faculty average; named to the **NUS Honour List of Student Tutors**.

Projects

Trader's Edge (Final-Year Project) | AI, Trading Strategy, Backtesting

May 2026 – Present

- Building an AI platform (advisor Anand Bhojan) that generates and backtests trading strategies and packages them into an interactive, data-rich learning environment; **working MVP shipped**.

Prediction-Market Arbitrage Engine | Python, EV Modelling

Feb 2026 – Present

- Prototyped a **cross-venue arbitrage model** for prediction markets (Polymarket, Kalshi), surfacing mispricings through expected-value and execution-cost modelling.

Model-Compression Pipeline for BERT QA | PyTorch, Quantisation, Pruning

Nov 2025

- Cut inference footprint by **69%** (440MB → 128MB) at **≤0.5% F1 cost** via post-training quantisation, with custom mask-enforced pruning to **54.7% sparsity**.

Honours

Top **0.5 percentile** in JEE (IIT entrance) out of **1.2 million** candidates · **National Topper in Mathematics**, Class 12 (3rd in State) · NUS Honour List of Student Tutors (2025).